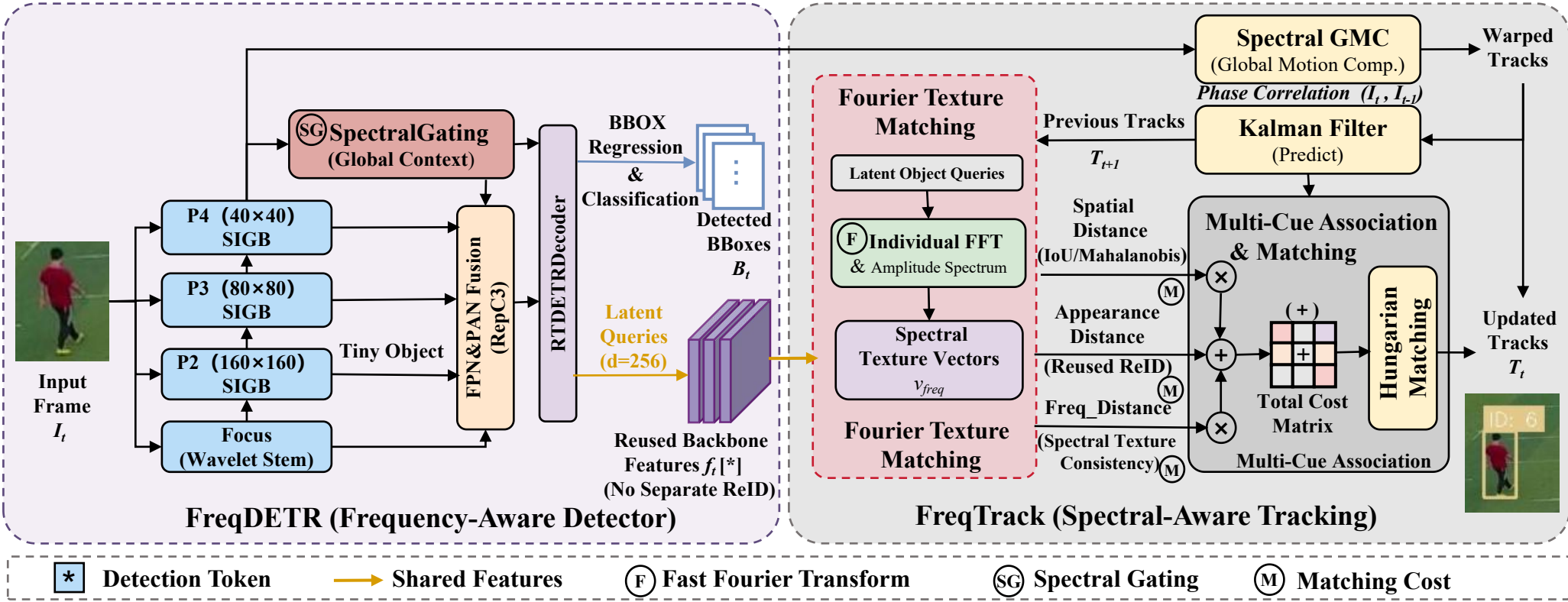
Input Frame I_t
P4 (40×40) SIGB
P3 (80×80) SIGB
P2 (160×160) SIGB
Focus (Wavelet Stem)
SG SpectralGating (Global Context)
Tiny Object
FPN&PAN Fusion (RepC3)
RTDETRDecoder
BBOX Regression & Classification
Detected BBoxes B_t
Latent Queries (d=256)
Reused Backbone Features f_t [*] (No Separate ReID)
FreqDETR (Frequency-Aware Detector)
Spectral GMC (Global Motion Comp.)
Warped Tracks
Phase Correlation (I_t, I_{t-1})
Kalman Filter (Predict)
Previous Tracks T_{t+1}
Fourier Texture Matching
Latent Object Queries
F Individual FFT & Amplitude Spectrum
Spectral Texture Vectors v_{freq}
Fourier Texture Matching
Spatial Distance (IoU/Mahalanobis)
Appearance Distance (Reused ReID)
Freq_Distance (Spectral Texture Consistency)
M
Multi-Cue Association & Matching
(+)
Total Cost Matrix
Hungarian Matching
Multi-Cue Association
Updated Tracks T_t
ID: 6
FreqTrack (Spectral-Aware Tracking)
* Detection Token
Shared Features
F Fast Fourier Transform
SG Spectral Gating
M Matching Cost